Safety Boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 31: permission is tied to a defined task.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of untrusted input boundaries. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **untrusted input** boundaries.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

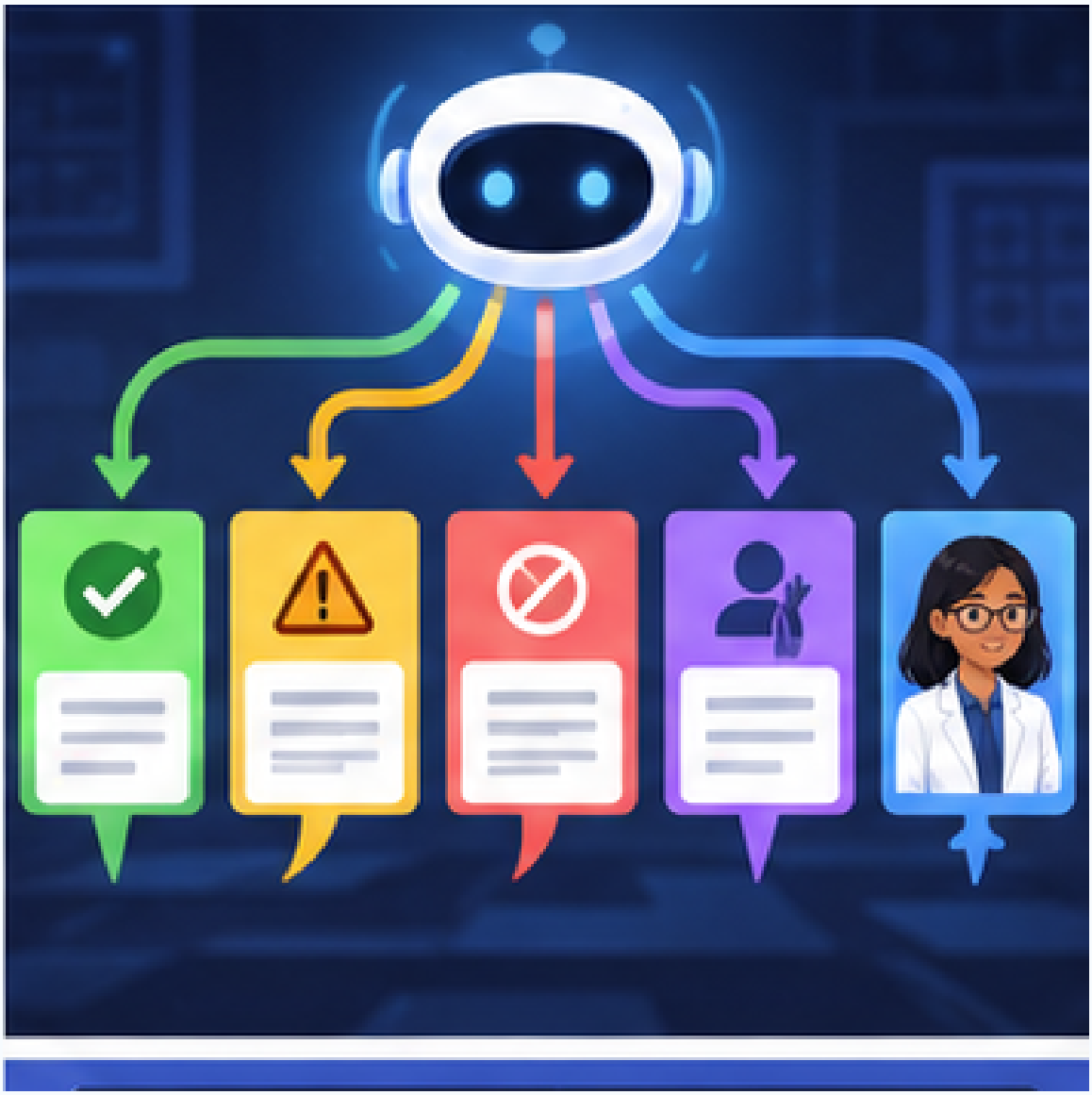
Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 31: Safety Boundaries

Systems need **refusal**, warning, **escalation**, and human-review policies.



CREATOR MISSION

Write safe behavior for three classroom risk levels.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall External side effects and recovery. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

Readiness for Abstention and clarification

Recall Claim support and citation evaluation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My readiness explanation**

This week’s end goal

By the end, I can explain safety boundaries using a traced example and a stated limitation.

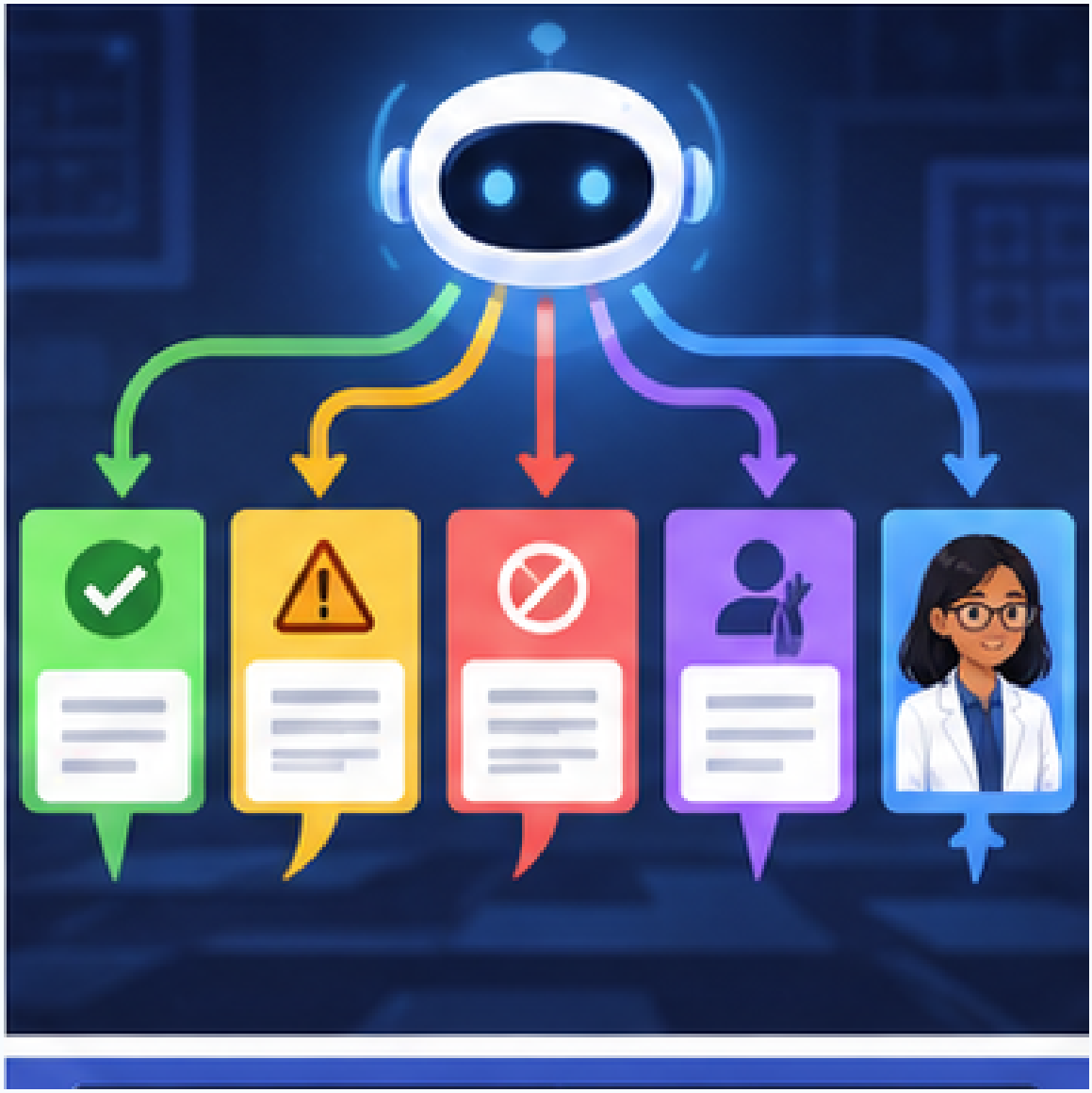
Notice the evidence

> Investigate the supplied Safety Boundaries case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use safety boundaries to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Untrusted input must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Abstention and clarification

Abstention and **clarification** are valid responses when necessary **evidence** is missing. Ask for the smallest fact that can resolve the decision.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

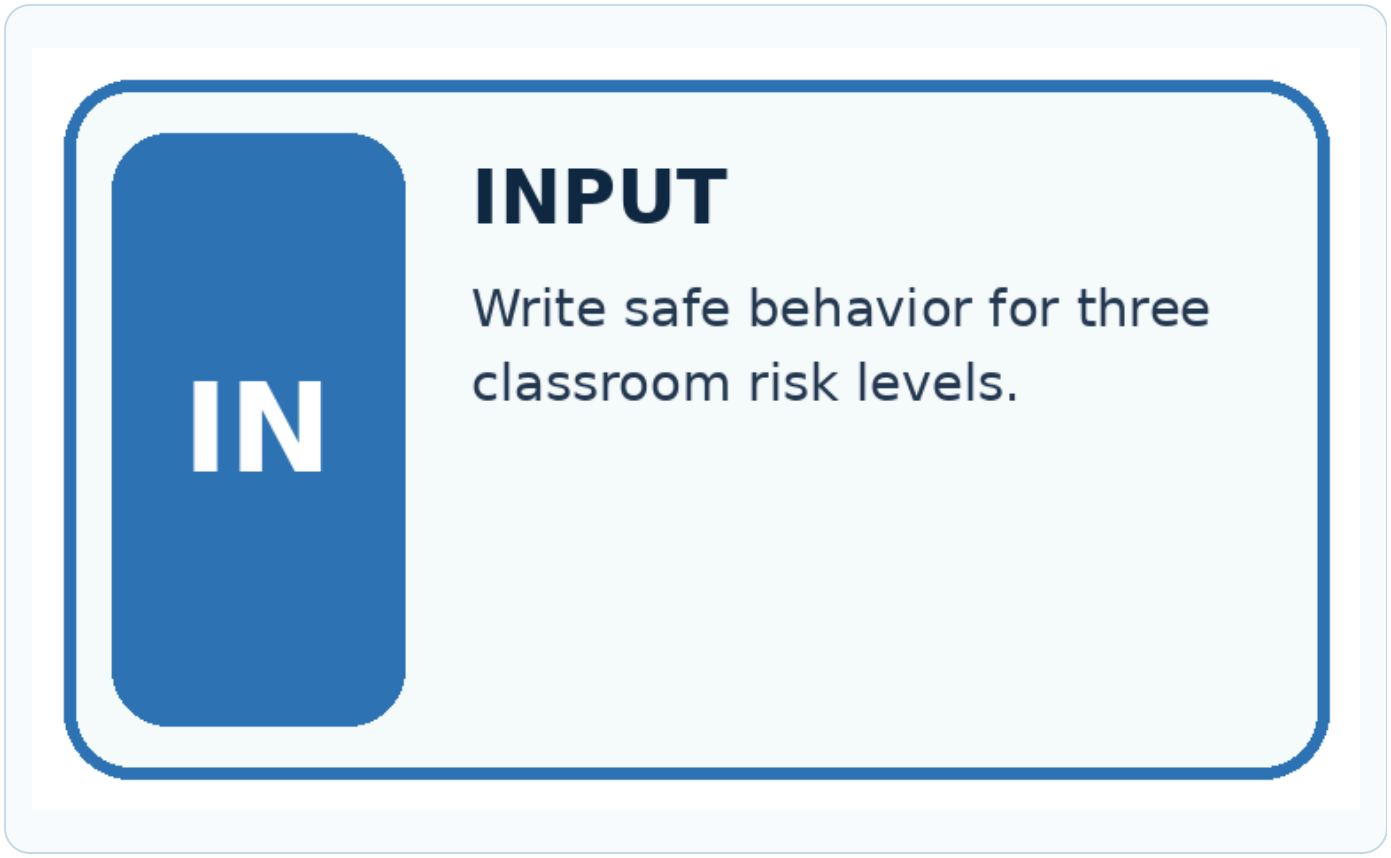
See how it works

> Follow the mechanism for Safety Boundaries. Connect each labelled input, deciding step and result.

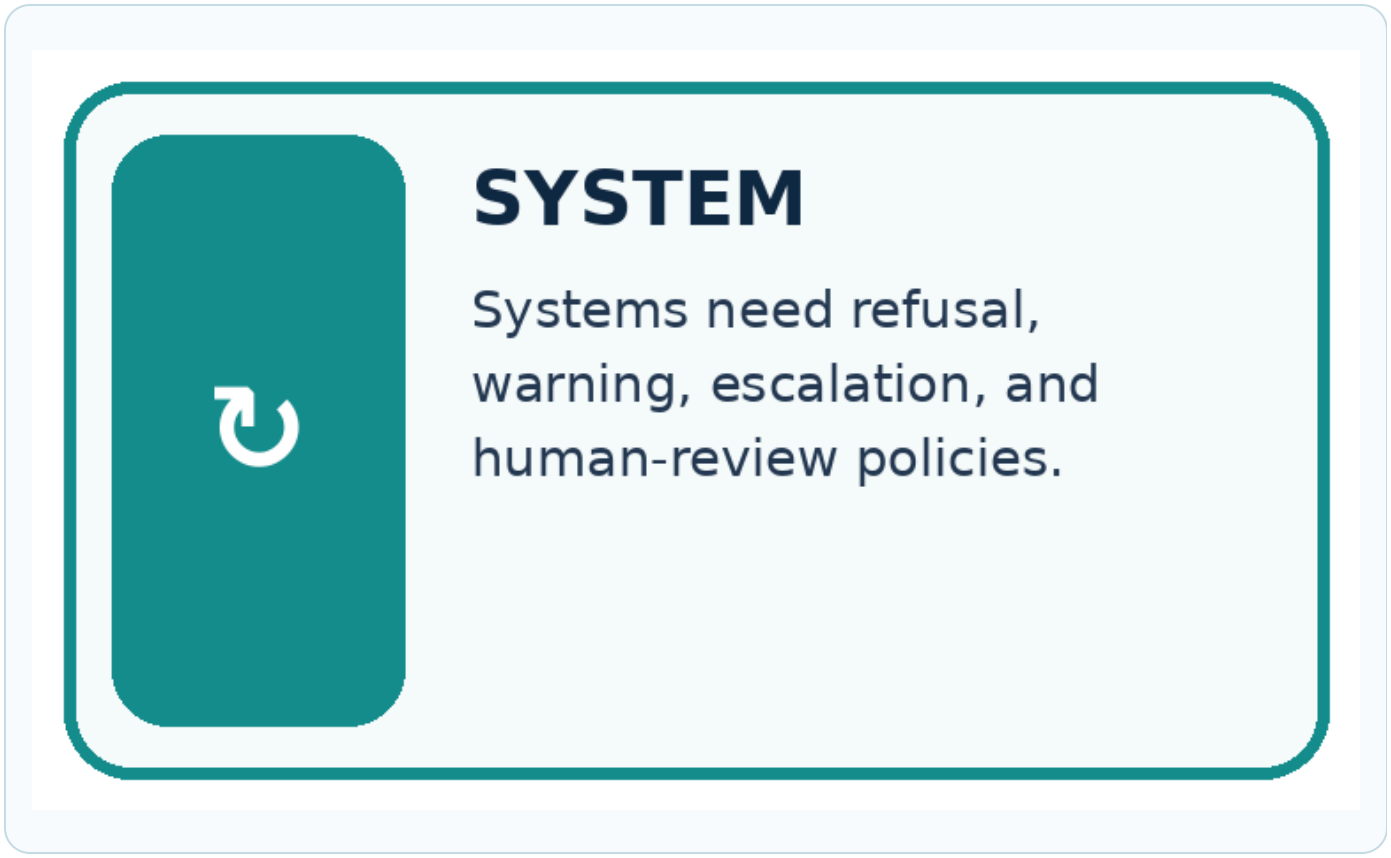
VISUAL MECHANISM

Input → Process → Output

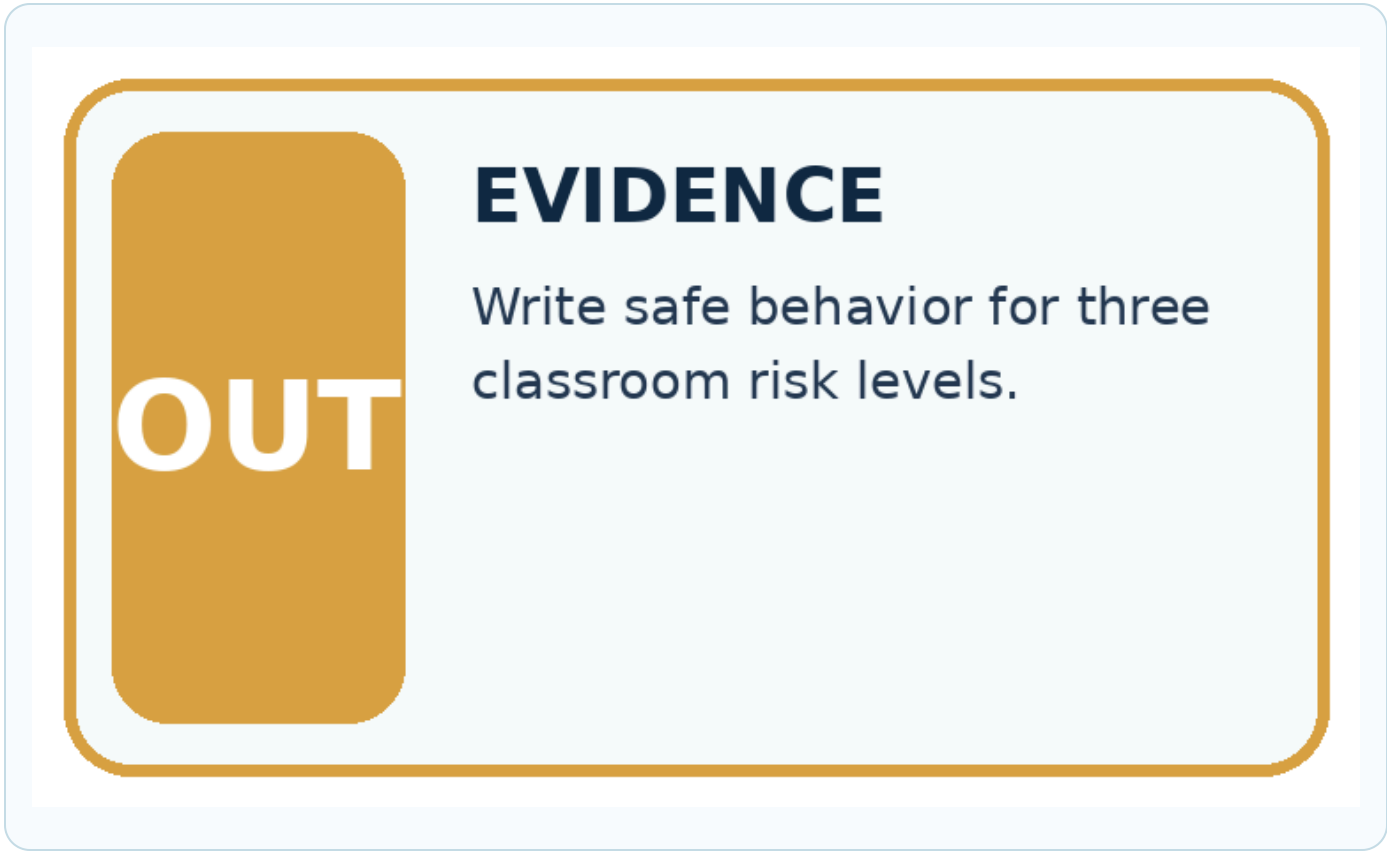
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Write safe behavior for three classroom risk levels.

↓


SYSTEM
Systems need refusal, warning, escalation, and human-review policies.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

Untrusted input boundaries — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Inputs A1 and delete_all() are supplied to room lookup.	For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.	A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision **boundary** where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Abstention and clarification

Separate answerable claims from unresolved ones. Clarify ambiguous **input** when a user can supply the needed fact; abstain from unsupported conclusions when **evidence** is unavailable. Define acceptable fallback before evaluation.

Abstention and clarification — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Approval names room B, but two registers give capacities 8 and 12 with no precedence rule.	Separate answerable claims from unresolved ones. Clarify ambiguous input when a user can supply the needed fact; abstain from unsupported conclusions when evidence is unavailable. Define acceptable fallback before evaluation.	Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

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If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

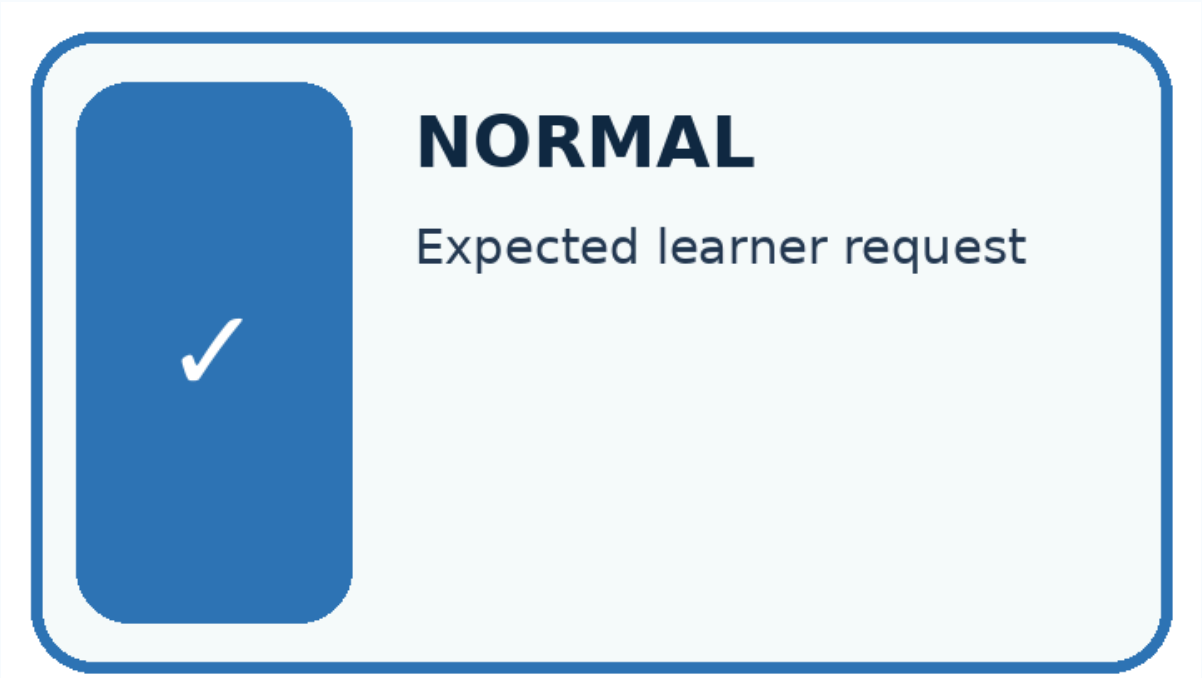
> Compare the supplied cases for Safety Boundaries. Identify the detail that changes the conclusion.

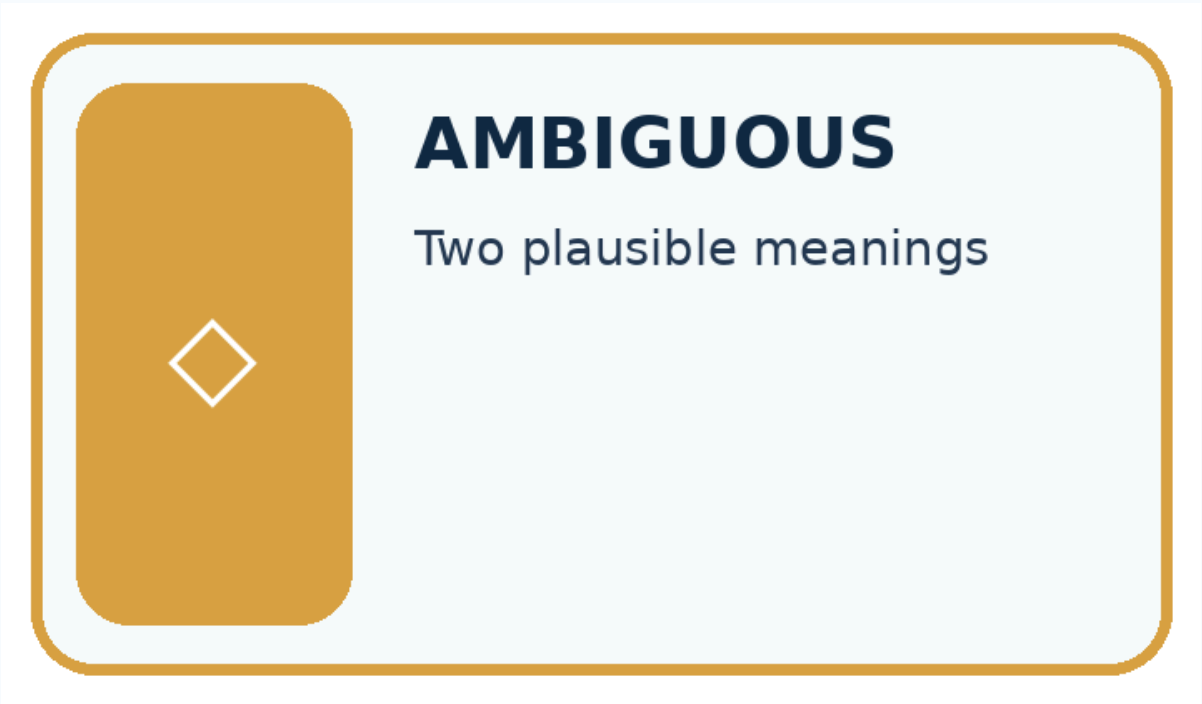
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

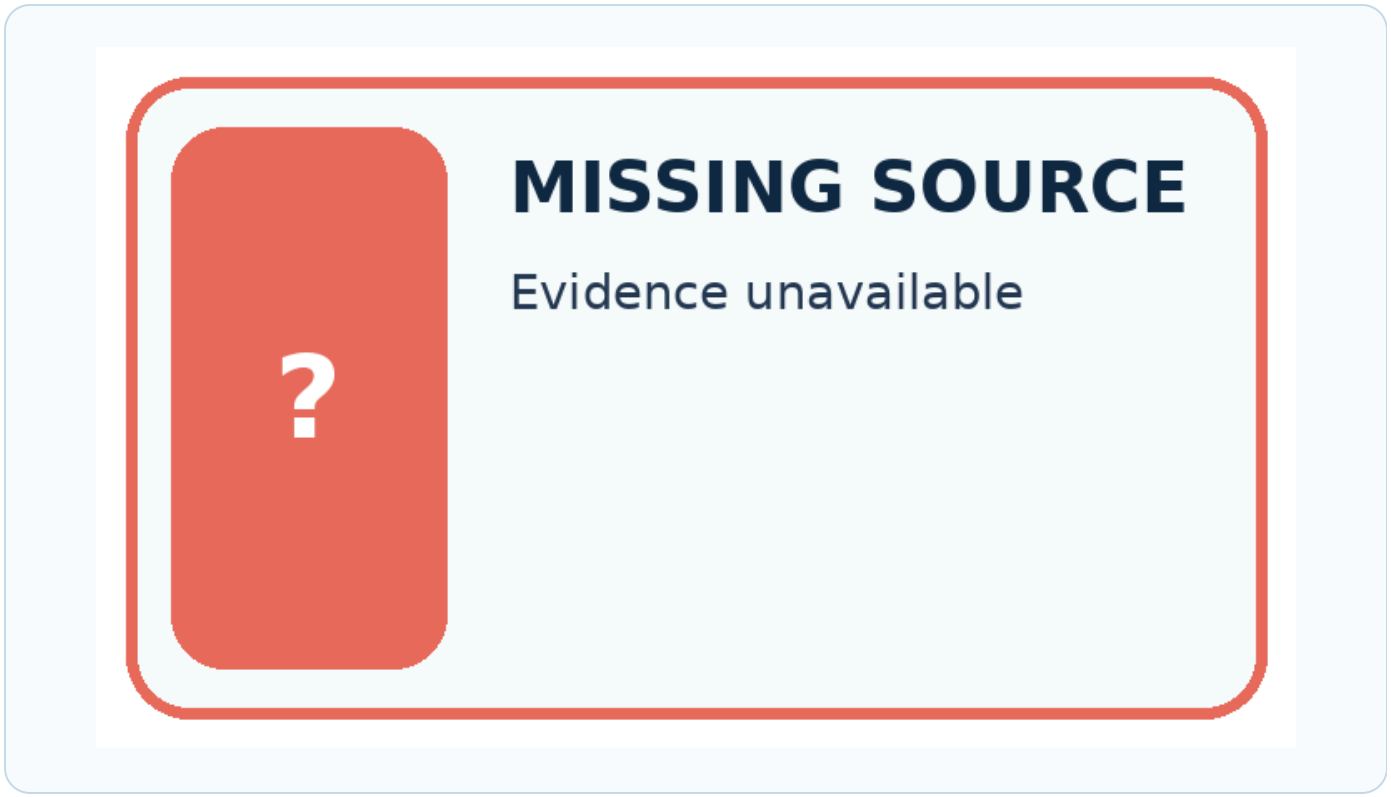
A system is not understood until it survives more than one comfortable example.

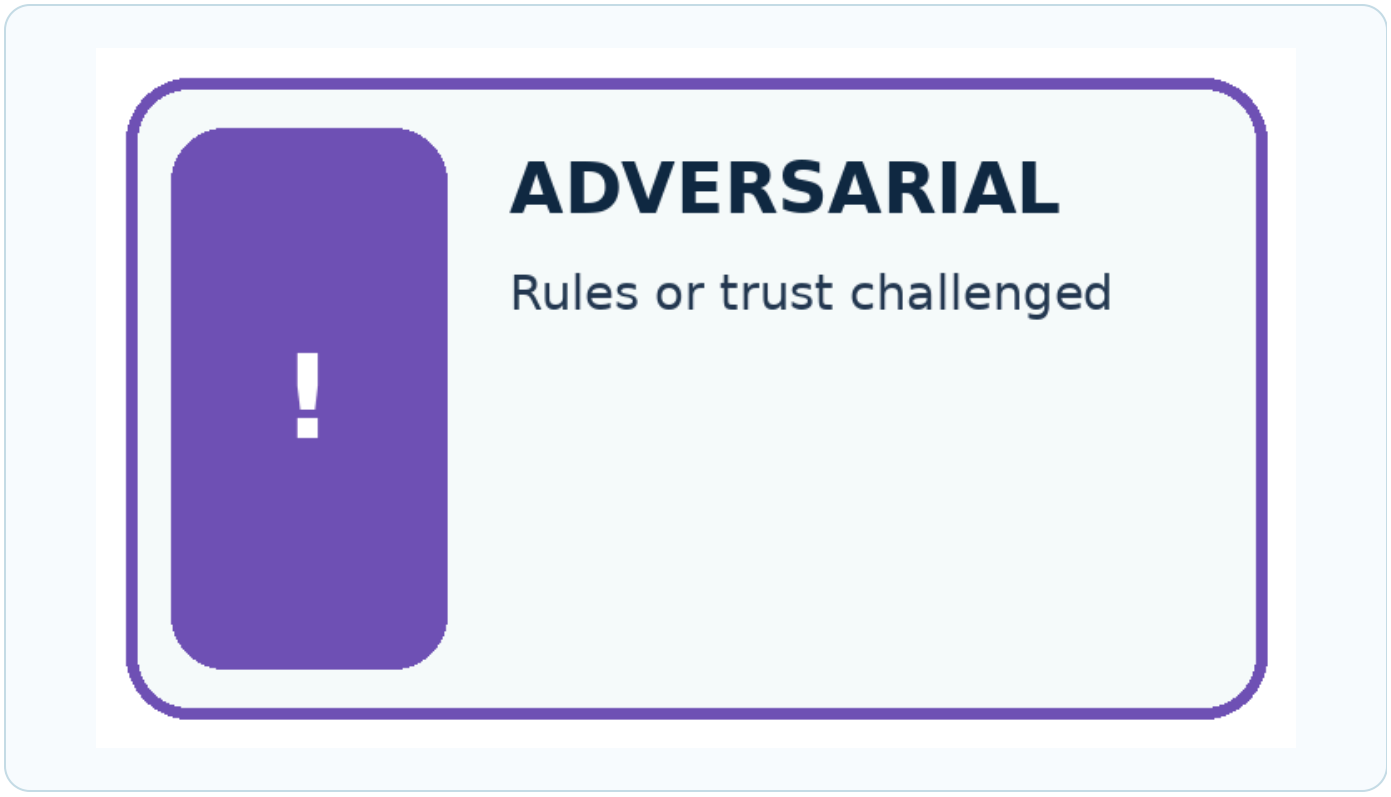




The normal case

NORMAL CASE: predict and name **evidence**. **BOUNDARY** CASE: predict and name **evidence**.


?
MISSING SOURCE
Evidence unavailable


!
ADVERSARIAL
Rules or trust challenged

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Change AB7 to Ab7.

Predicted result

Prediction: Reject it under the exact uppercase rule.

Why the result follows

Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

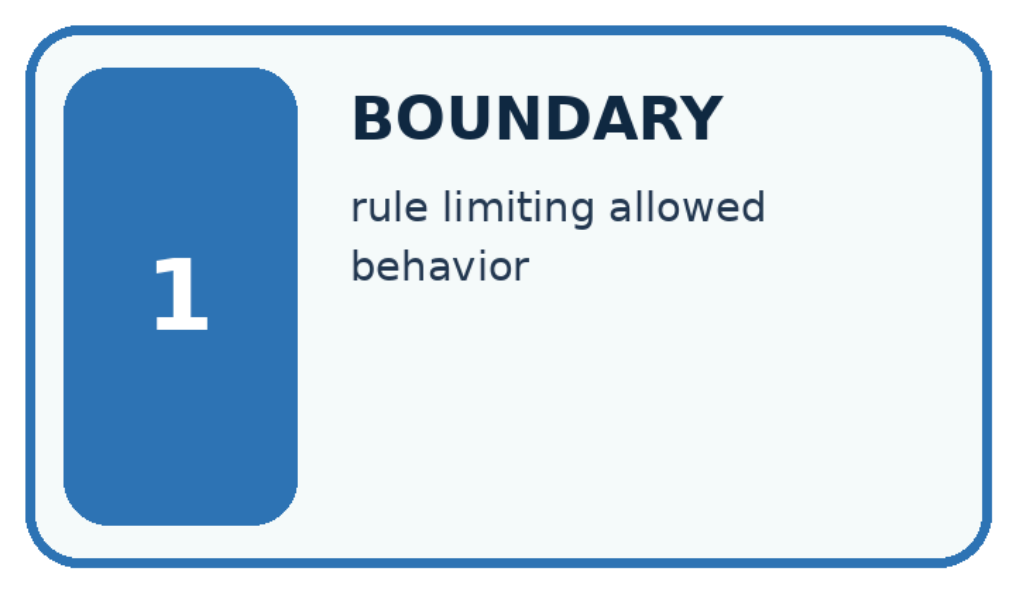
Words that unlock the idea

> Use the vocabulary of Safety Boundaries to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
REFUSAL
decline a disallowed request
3
ESCALATION
route to an authorized human



Word	My own example	Not this / boundary
boundary	A dividing condition between categories, actions or allowed behaviours.	
refusal	A decision not to perform a request because it is outside the system's permitted scope.	
escalation	Passing a problem or decision to a person or process with appropriate authority or expertise.	
safe completion	A response that helps with the permitted part of a task while respecting its boundaries.	

Words used in the closer explanation

Word	Meaning
Untrusted input	Data whose claims or control intent have not been authorised
Full match	Validation covering the whole input
Injection boundary	A separation preventing data from becoming executable control

Terms for Abstention and clarification

Term	Meaning
Abstention	Declining to state an unsupported conclusion
Clarification	A targeted request for missing or ambiguous information
Answerability	Whether supplied evidence can support the requested conclusion

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Safety Boundaries, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 31 • WORKED EVIDENCE

1. Low risk: answer a source-grounded study question.
2. Uncertain risk: warn, ask for context, or request adult help.
3. Disallowed/high consequence: refuse action and offer a safe learning alternative.
4. Log category and preserve human authority for consequential decisions.

1. Low risk: answer a source-grounded study question.

2. Uncertain risk: warn, ask for context, or request adult help.

3. Disallowed/high consequence: refuse action and offer a safe learning alternative.

4. Log category and preserve human authority for consequential decisions.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Inputs A1 and delete_all() are supplied to room lookup.

A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Worked explanation · Safety Boundaries

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Safety Boundaries

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant **boundary input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
import re

def validate_identifier(value):
    return type(value) is str and re.fullmatch('[A-Z][0-9]', value) is not None

print(validate_identifier("B2"))
```

Apply the connected idea: Abstention and clarification

Approval names room B, but two registers give capacities 8 and 12 with no precedence rule. The question asks maximum attendance.

Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

> ### Worked explanation · Safety Boundaries
>
> Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.
>
> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Safety Boundaries

Make a strict lookup identifier **boundary** visible.

The facts we will use

Supplied facts

Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/31

Safety Boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check pattern

AB7 has two permitted letters and one digit, with no extra characters.





2. Reject extra text

AB7; send() violates the exact identifier pattern.





3. Separate authority

Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

CHANGE ONE THING

Change AB7 to Ab7.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check pattern	AB7 has two permitted letters and one digit, with no extra characters.
2. Reject extra text	AB7; send() violates the exact identifier pattern.
3. Separate authority	Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.

Keep the stated limit

Read the labelled picture: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Change AB7 to Ab7. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Safety Boundaries claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A safe system refuses anything difficult or unfamiliar. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

Apply the connected idea: Abstention and clarification

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

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STUDENT LESSON · 8 OF 18

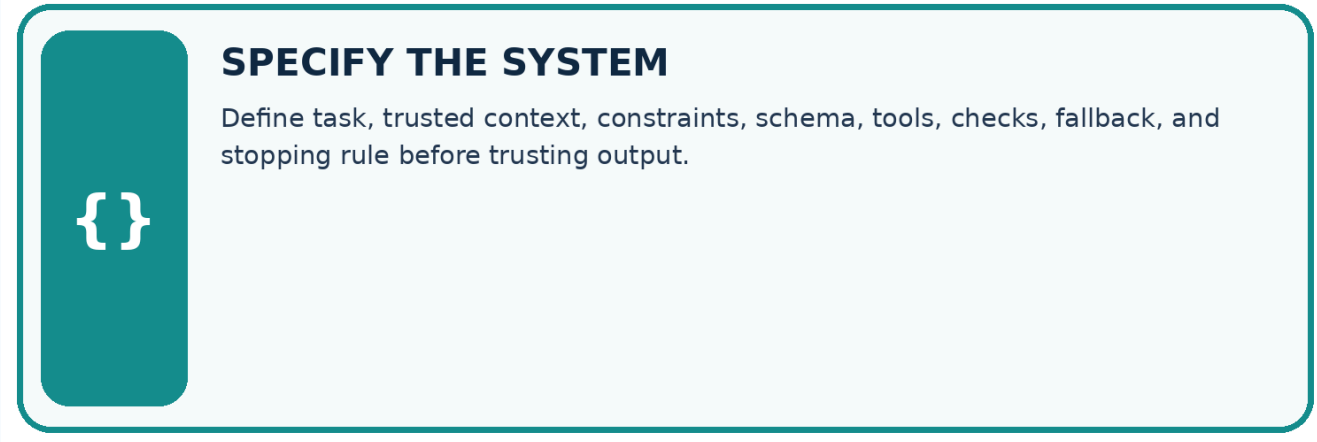
Practise together

> Practise Safety Boundaries with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A format check does not grant resource access; authorisation follows separately.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Apply the connected idea: Abstention and clarification

An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Untrusted input** boundaries". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Untrusted input** boundaries. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Abstention and clarification

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Explain the picture

If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Safety Boundaries. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Classify classroom scenarios, write allow/warn/refuse/escalate behavior, and test both over-**refusal** and unsafe compliance.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

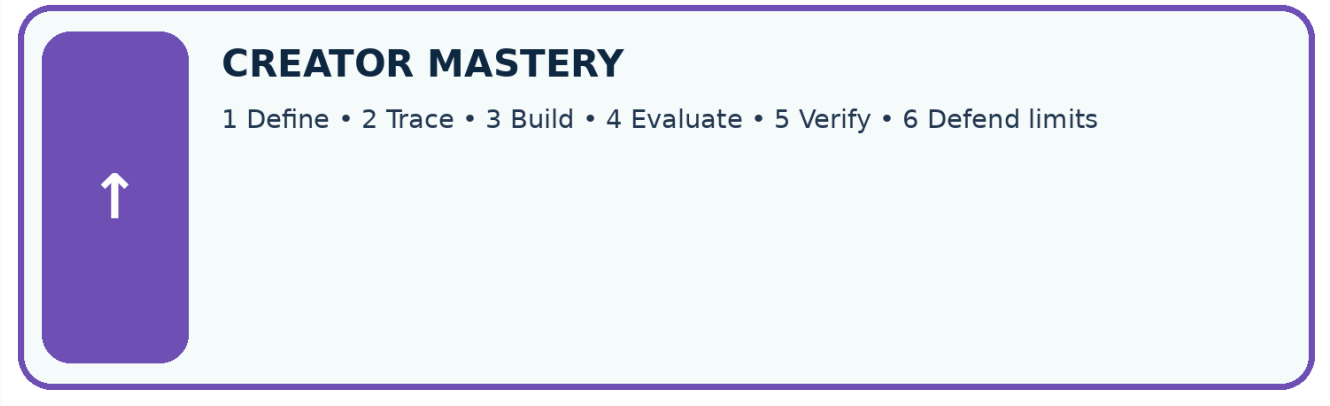
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Safety Boundaries. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a **boundary** case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

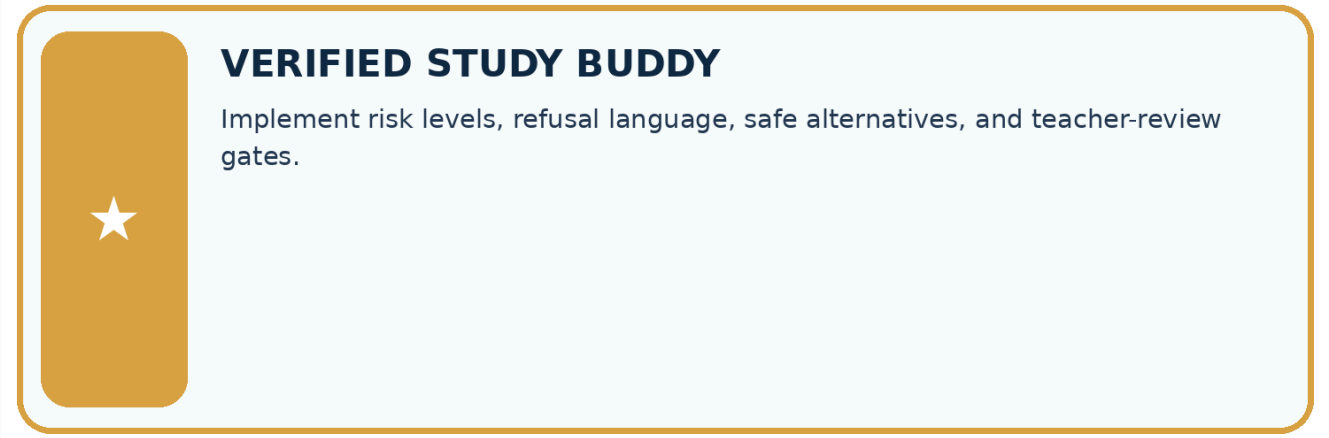
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Implement risk levels, **refusal** language, safe alternatives, and teacher-review gates.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

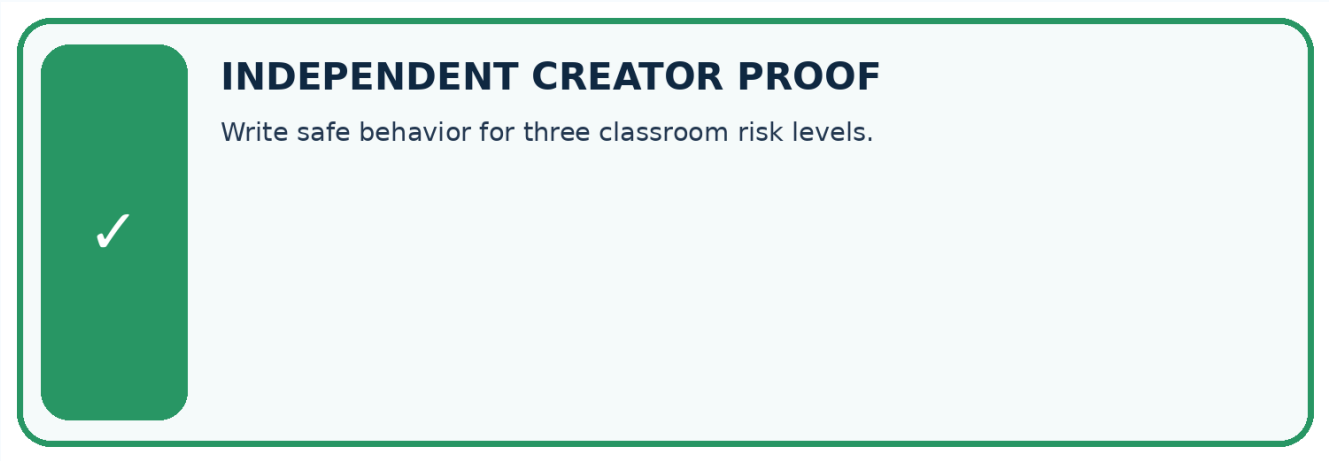
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Write safe behavior for three classroom risk levels.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a **boundary** or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Safety Boundaries case for supported practice. Repeated examples do not become fresh assessments.

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A Study Buddy is asked to decide whether a learner should take medicine.

The question

Set a product **boundary**.

Worked explanation

A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Keep a study tool inside its role

Evidence or stage	Value or result
Study explanation	Approved educational content
Individual treatment	Outside this product's role

Evidence or stage	Value or result
Response	Direct to appropriate human help

A Study Buddy is asked to decide whether a learner should take medicine.

Set a product **boundary**.

A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Classify B2, b2, B22 and B2; DROP under the exact format.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 31 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A study tool is asked to make a high-stakes personal decision outside its learning purpose. What **boundary** should the response show?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a **boundary** or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Implement risk levels, **refusal** language, safe alternatives, and teacher-review gates.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A plant-study assistant is asked for a person's medication dose. Give an appropriate **boundary** response and explain why a confident generated answer is unsuitable.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Safety Boundaries, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Make a strict lookup identifier **boundary** visible.

By the end, I can	How I will show it
Explain the mechanism and terminology of untrusted input boundaries	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Safety Boundaries without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **untrusted input** boundaries

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Make a strict lookup identifier **boundary** visible.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

1. Separate authority

2. Check pattern

3. Reject extra text

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Separate authority Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Read the picture in order.

Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change AB7 to Ab7. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Reject it under the exact uppercase rule.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject it under the exact uppercase rule. Lowercase b fails the declared pattern even though the identifier looks similar.

Repair a missing building block

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Quality Rubrics (ai-3/25). Goal: Explain the **mechanism** and terminology of system evaluation sets. Short practice: Why is “one out of two checks passed” an insufficient release decision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

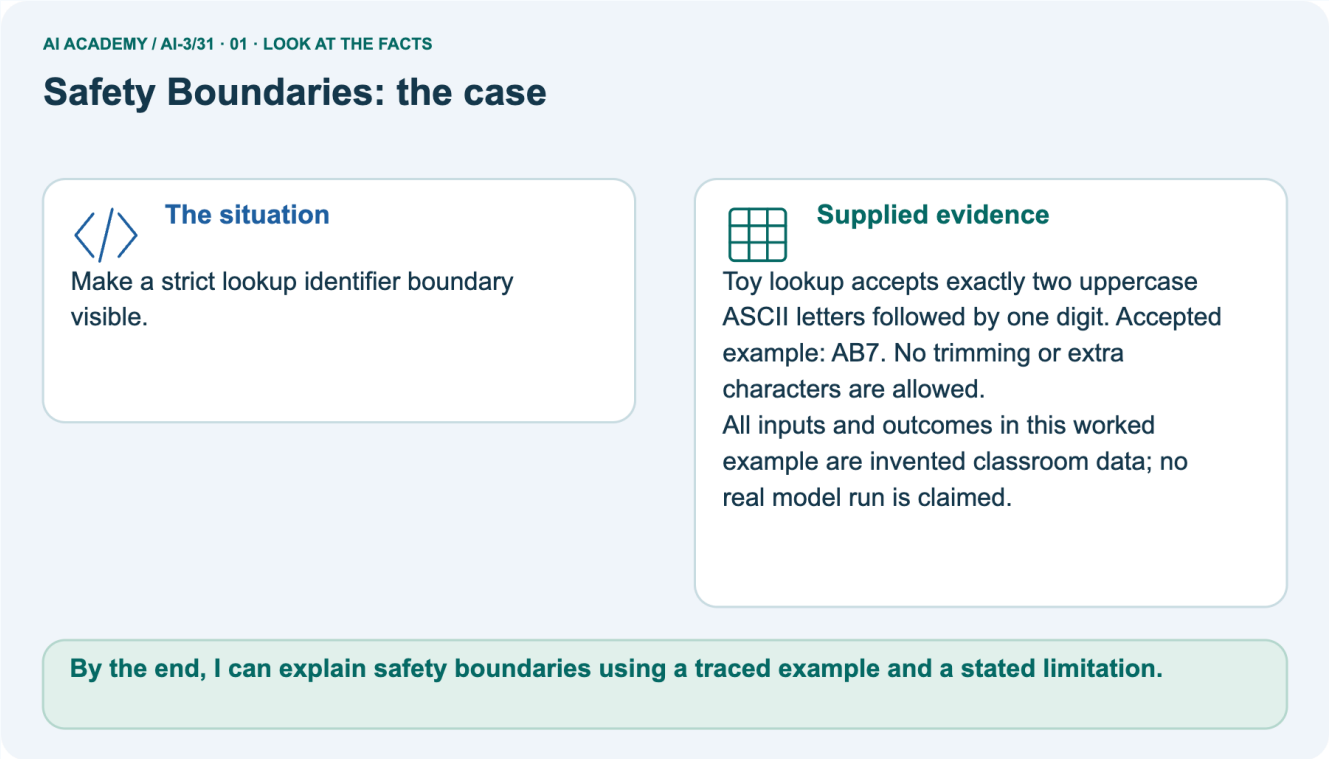
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Safety Boundaries graphic using its labels, deciding evidence and relevant limitation.

01. Safety Boundaries: the case



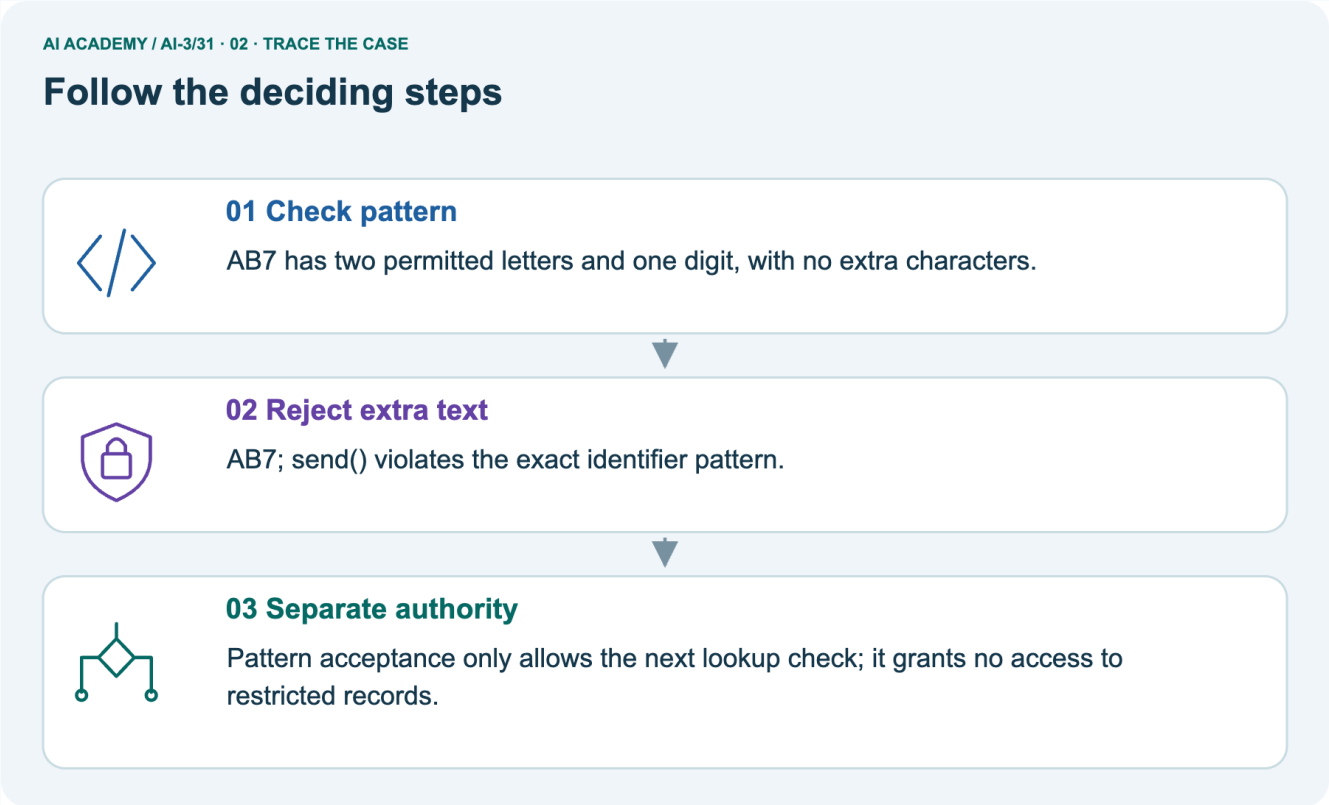
Follow the labelled picture with the matching explanation. Open the vector diagram

Make a strict lookup identifier **boundary** visible. Supplied facts: Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Check pattern: AB7 has two permitted letters and one digit, with no extra characters. → Reject extra text: AB7; send() violates the exact identifier pattern. → Separate authority: Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Untrusted input boundaries — evidence trace

AI ACADEMY / AI-3/31 · 03 · INSPECT THE LESSON MODEL

Untrusted input boundaries — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Inputs A1 and delete_all() are supplied to room lookup.	For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.	A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

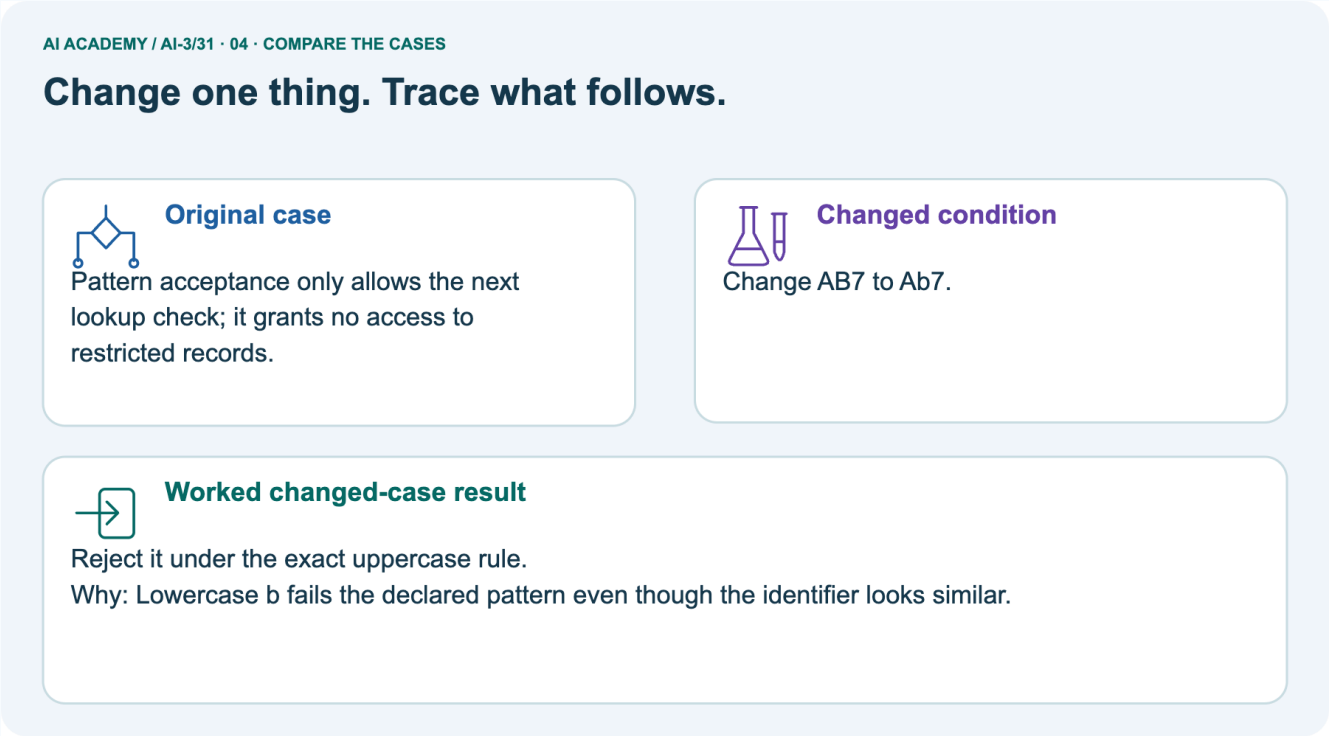
Follow the labelled picture with the matching explanation. Open the vector diagram

Untrusted input boundaries — **evidence** trace. **Input**s A1 and delete_all() are supplied to room lookup. | For the toy lookup accept room IDs matching one uppercase letter plus one digit.

Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters. | A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



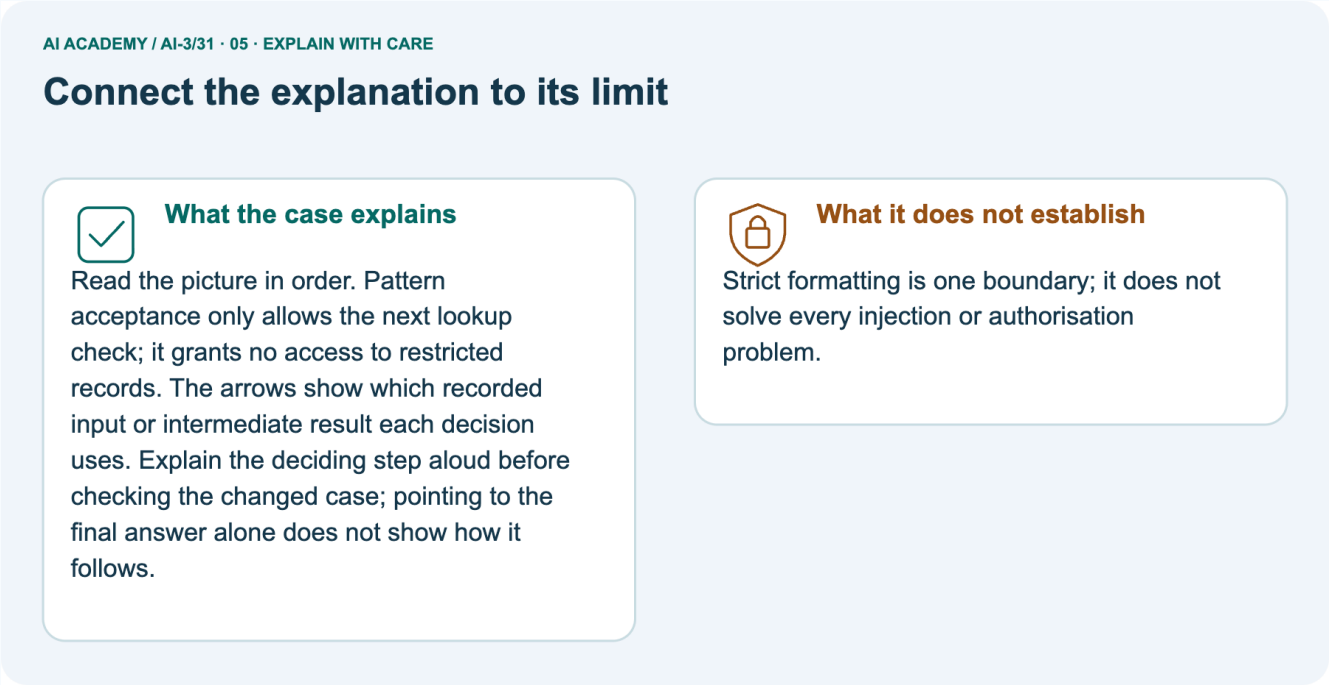
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Changed condition: Change AB7 to Ab7. Expected result: Reject it under the exact uppercase rule.

Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

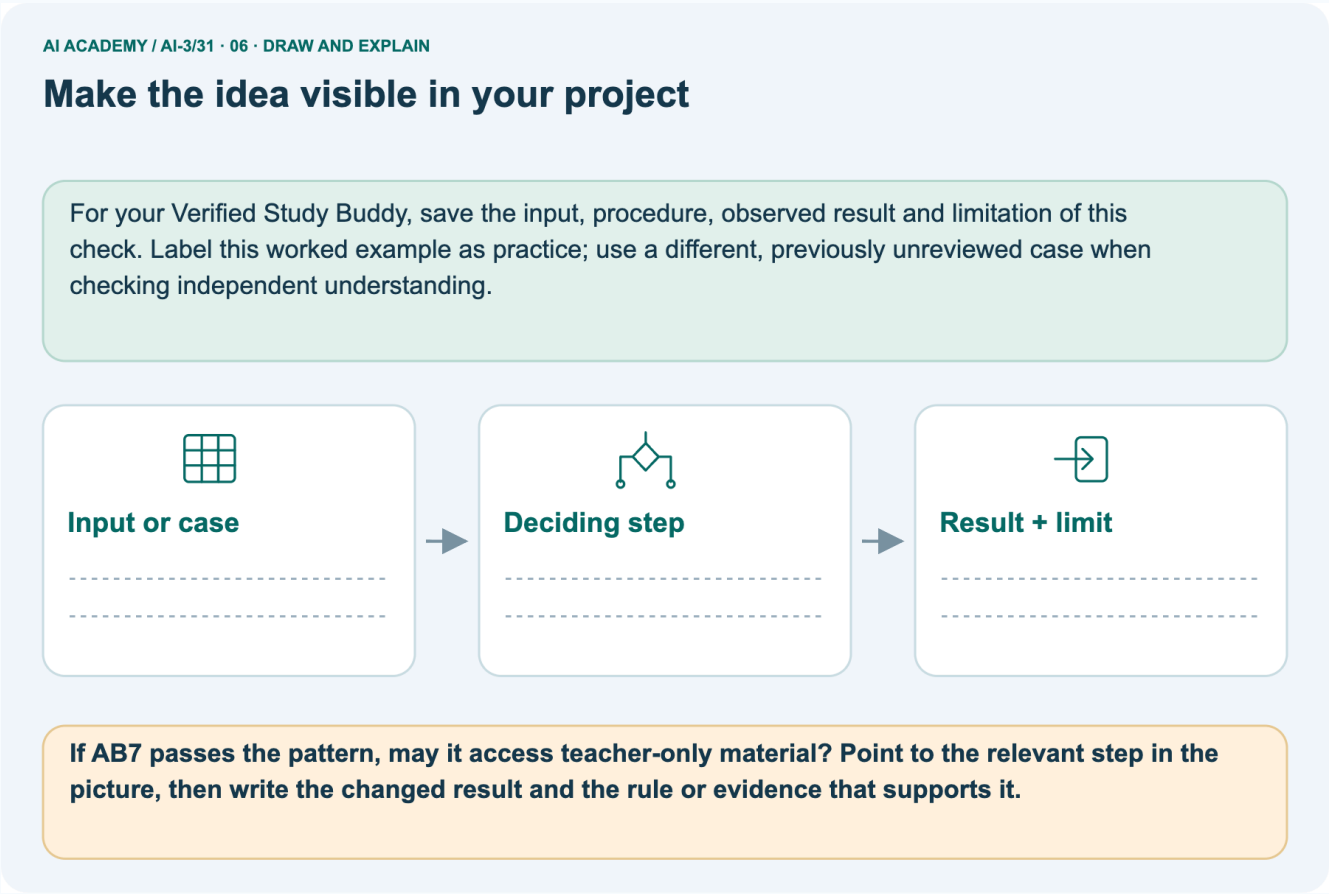
Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.